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Communication Connector

Abstract

A communication connector includes a housing, a first conducting portion, a second conducting portion, and a third conducting portion. The housing is composed of an electrically conductive material and receives a mating connector. The first conducting portion is retained in the housing and conductively connected to the housing. The first conducting portion is conductively connected to a shield member of the mating connector. The second conducting portion is provided on one side of the housing and is conductively connected to the housing. The third conducting portion is provided on another side of the housing and is conductively connected to the housing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of the filing date under 35 U.S.C. § 119(a)-(d) of Japanese Patent Application No. JP2024-028398, filed on Feb. 28, 2024.

FIELD OF THE INVENTION

[0002] The present invention relates to a connector and, more particularly, to a communication connector.

BACKGROUND OF THE INVENTION

[0003] Communication connectors are required to reduce the effect of electromagnetic noise on communication signals to ensure good communication characteristics. A communication connector, in one example, is mated with a mating connector while being connected to a circuit board accommodated inside an electrically conductive enclosure. A communication connector disclosed in Japanese Patent Application No. JP2023-102172A, for example, includes a contact molded body including two contacts, and a resin housing accommodating an electrically conductive tubular member. While reducing the effect of electronic noise, however, the structure of the communication connector should avoid being complicated.

SUMMARY OF THE INVENTION

[0004] A communication connector includes a housing, a first conducting portion, a second conducting portion, and a third conducting portion. The housing is composed of an electrically conductive material and receives a mating connector. The first conducting portion is retained in the housing and conductively connected to the housing. The first conducting portion is conductively connected to a shield member of the mating connector. The second conducting portion is provided on one side of the housing and is conductively connected to the housing. The third conducting portion is provided on another side of the housing and is conductively connected to the housing.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0005] The invention will now be described by way of example with reference to the accompanying figures, of which:

[0006] FIG. **1** is an isometric view of a communication connector and a mating connector according to an exemplary embodiment;

[0007] FIG. **2** is a cross-sectional view of the communication connector and the mating connector of FIG. **1** taken along line II-II of FIG. **1**;

[0008] FIG. **3** is an isometric view of the communication connector of FIG. **1** with a ground piece detached therefrom;

[0009] FIG. **4** is an isometric view of the communication connector of FIG. **1**;

[0010] FIG. **5** is a cross-sectional view of the communication connector of FIG. **1** taken along line V-V of FIG. **4**;

[0011] FIG. **6** is an isometric view of a communication connector according to another exemplary embodiment;

[0012] FIG. **7** is a cross-sectional view of the communication connector of FIG. **6** taken along line VII-VII of FIG. **6**;

[0013] FIG. **8** is an isometric view of a communication connector according to another exemplary embodiment;

[0014] FIG. **9** is a cross-sectional view of the communication connector of FIG. **8** taken along line IX-IX of FIG. **8**;

[0015] FIG. **10**A is an isometric view of a communication connector according to another exemplary embodiment;
[0016] FIG. **10**B is a detail side view of a portion of the communication connector of FIG. **10**A, taken along arrow (b) in FIG. **10**A;
[0017] FIG. **10**C is a detail sectional side view of the portion of the communication connector of FIG. **10**B, taken along line (c)-(c) of FIG. **10**B;
[0018] FIG. **11** is a cross-sectional view of the communication connector of FIG. **10**A taken along line XI-XI of FIG. **10**A; and
[0019] FIG. **12** is an isometric view of a communication connector according to another exemplary embodiment.

DETAILED DESCRIPTION

[0020] A communication connector **1** according to each exemplary embodiment of the present invention will be described below with reference to the drawings. It should be noted that the term "connection" as used herein means electrical connection and mechanical connection.

[0021] It should be noted that, for descriptive convenience, as shown in each drawing, a mating direction of the communication connector **1** and a mating connector **100** is a first direction X, a direction orthogonal to the first direction X is a second direction Y, and a direction orthogonal to the second direction Y is a third direction Z. In addition, (F) indicates frontward in the first direction X, (B) indicates rearward in the first direction X, (R) indicates rightward in the second direction Y, (L) indicates leftward in the second direction Y, (U) indicates upward in the third direction Z, and (D) indicates downward in the third direction Z.

[0022] An exemplary embodiment of the communication connector **1** according to FIGS. **1-5** will now be described. As shown in FIG. **5**, the communication connector **1** is mounted on a circuit board **200** installed inside an enclosure **300**. The communication connector **1** is mated with a mating connector **100**, as shown in FIGS. **1-3** and **5**. As shown in FIG. **5**, the communication connector **1** mated with the mating connector **100** is inserted through an opening portion **301** of the enclosure **300**, thereby relaying communication signals from the mating connector **100** to the circuit board **200**.

[0023] As shown in FIGS. **1-5**, the communication connector **1** includes a housing **10**. The housing **10** receives the mating connector **100**, as shown in FIGS. **1-3** and **5**, when the communication connector **1** and the mating connector **100** are mated. As shown in FIG. **2**, the communication connector **1** also includes a tubular member **30** as a first conducting portion conductively connected to a shield member **110** of the mating connector **100**. As shown in FIG. **5**, the communication connector **1** also includes a ground piece **40**A as a second conducting portion conductively connected to the enclosure **300**. As shown in FIG. **2**, the communication connector **1** also includes a contact molded body **50** provided with a contact portion **51** brought into contact with a contact portion **130** of the mating connector **100** to relay differential signals.

[0024] The housing **10** is formed with a mating opening **11**, as shown in FIGS. **2** and **5**, a catching hole **12**, as shown in FIGS. **1-2**, **4**, and **5**, a retaining portion **13**, as shown in FIGS. **2** and **5**, an attachment portion **20**A, as shown in FIGS. **2** and **4-5**, and a lead portion **15**, as shown in FIGS. **2** and **4-5**.

[0025] As shown in FIG. **2**, a housing **150** of the mating connector **100** is inserted through the mating opening **11**, and the catching hole **12** connects with the mating opening **11** and catches a catching portion **140** of the mating connector **100**. The retaining portion **13** retains the tubular member **30** as the first conducting portion conductively connected to the shield member **110** of the mating connector **100**, as shown in FIG. **2**. The retaining portion **13** is conductively connected to the tubular member **30**. As shown in FIG. **5**, the attachment portion **20**A elastically deforms, thereby pressing the enclosure **300**. The ground piece **40**A as the second conducting portion conductively connected to the enclosure **300** is attached to the attachment portion **20**A, as shown in FIG. **5**. As shown in FIG. **5**, the lead portion **15** is a third conducting portion inserted into the circuit board **200**. The tubular member **30** as the first conducting portion, the ground piece **40**A as the second conducting portion, and the lead portion **15** as the third conducting portion are conductively connected to each other.

[0026] The housing **150** of the mating connector **100**, at the time of mating with the mating connector **100**, is inserted into the housing **10**. The housing **10** thus receives the mating connector **100**. As shown in FIG. **5**, a front face **10**F of the housing **10** in the first direction X receives the mating connector **100**.

[0027] In the present embodiment, the housing **10** is composed of an electrically conductive material. For example, the housing **10** is composed of a die-cast member composed of an aluminum alloy.

[0028] As shown in FIGS. **2** and **5**, the mating opening **11** is formed at the front face **10**F of the housing **10**, through which the mating connector **100** is inserted. As shown in FIG. **4**, the catching hole **12** is formed in an upper face **10**U of the housing **10**. The catching hole **12** connects with the mating opening **11**, as shown in FIG. **5**. As shown in FIG. **2**, when the mating connector **100** is inserted through the mating opening **11**, the catching portion **140** of the mating connector **100** is caught in the catching hole **12**, thus preventing unexpected extraction of the mating connector **100**.

[0029] As shown in FIGS. **2** and **5**, the retaining portion **13** retains the tubular member **30**. The retaining portion **13** is conductively connected to the tubular member **30** through a retained part of the tubular member **30**.

[0030] As shown in FIG. **2**, the contact molded body **50** is inserted into the tubular member **30**. The tubular member **30** has the function of shielding an inserted portion of the contact molded body **50**. As further shown in FIG. **2**, the tubular member **30** is conductively connected to the shield member **110** on an outer peripheral face of the tubular member **30**.

[0031] The contact molded body **50** is a component composed of the contact portion **51**, and a resin body **52** integrally molded with the contact portion **51**, as shown in FIG. **5**. The contact portion **51** is composed of an electrically conductive member and, as shown in FIG. **2**, has the function of coming into contact with a contact portion **130** of the mating connector **100** inserted through the mating opening **11** and relaying differential signals thereto. The contact portion **51** is bent in the middle to have a section extending in the first direction X and a section extending in the third direction Z.

[0032] As shown in FIG. **2**, a section of the contact portion **51** extending in the frontward/rearward direction is referred to as a first contact section **51***a*, the section of the contact portion **51** extending in the third direction Z is referred to as a second contact section **51***b*, and a further section extending in the rearward direction B from a lower end of the second contact section **51***b* is referred to as a third contact section **51***c*. As shown in FIG. **5**, the third contact section **51***c* is soldered to a contact portion of the circuit board **200** when the communication connector **1** is mounted on the circuit board **200**. That is, the third contact section **51***c* is surface-mounted on the circuit board **200**.

[0033] As shown in FIG. **4**, the attachment portion **20**A is formed in the upper face **10**U at the rear of the housing **10**. Specifically, as shown in FIGS. **4-5**, an attachment opening **22**A extending in the frontward direction F from a rear face **10**B of the housing **10** is formed. As shown in FIG. **5**, an insertion opening **23**A connecting with the attachment opening **22**A is formed in the upper face **10**U of the housing **10**. The attachment portion **20**A is formed with a latch portion **21**A, as shown in FIG. **5**.

[0034] As shown in FIGS. **3** and **5**, the ground piece **40**A, in a side view, is formed in a U-shape **41**A, and is formed with a shielding portion **43**A extending from an upper section **42**A of the U-shape **41**A, and a curved portion **44**A is formed on a distal end side of the shielding portion **43**A. A lower section **47**A of the U-shape **41**A extends in the frontward direction F, and extends obliquely downward in the frontward direction F on a distal end side. In addition, the shielding portion **43**A is formed with a notched portion **45**A, as shown in FIGS. **3-4**, thereby being divided into two branches. In addition, an opening **46**A is formed in a middle section of the U-shape **41**A. As shown in FIGS. **2** and **4-5**, when the ground piece **40**A is attached to the attachment portion **20**A, the latch portion **21**A of the attachment portion **20**A enters the opening **46**A to prevent unexpected extraction of the ground piece **40**A. The ground piece **40**A has a function as a flat spring for pressing the enclosure **300** with the curved portion **44**A by elastic deformation of a portion connecting from the upper section **42**A to the shielding portion **43**A and of the shielding portion **43**A.

[0035] As shown in FIGS. **2** and **4-5**, the lead portion **15** extends in the downward direction D from a lower face of the housing **10**. The lead portion **15** has a cylindrical shape, and includes five lead portions **15** formed on the housing **10**. When the communication connector **1** is mounted on the circuit board **200**, the lead portion **15** serves as a mount portion **15**A inserted into the circuit board **200** and further soldered thereto, and is conductively connected to the circuit board **200**. That is, after being inserted into the circuit board **200**, the cylindrical lead portion **15** is through-hole mounted and firmly secured.

[0036] As shown in FIG. **5**, each of the five lead portions **15** and the circuit board **200** are conductively connected to each other, thereby increasing the number of electric current paths for grounding. By forming a plurality of grounded electric current paths, the number of paths for releasing electric current that will cause noise to a portion where communication signals are not affected can be increased, and the occurrence of noise can be reduced. In addition, by forming the grounded electric current paths between the lead portions **15** and the circuit board **200**, an electric current conduction distance is shortened, and the occurrence of noise can be reduced.

[0037] As shown in FIG. **5**, the communication connector **1** is mounted on the circuit board **200** incorporated in the enclosure **300**. When the communication connector **1**, having the third contact section **51***c* conductively connected to the contact portion of the circuit board **200**, is mated with the mating connector **100**, the contact portion **130** of the mating connector **100** and the first contact section **51***a*, as shown in FIG. **2**, are brought into contact with each other, relaying signals send from the mating connector **100** to the circuit board **200**.

[0038] When the communication connector **1**, having the lead portion **15** conductively connected to a through-hole portion of the circuit board **200**, is mated with the mating connector **100**, the shield member **110** of the mating connector **100** and the tubular member **30** are conductively connected to each other. In addition, since the tubular member **30** and the retaining portion **13** retaining the tubular member **30** are conductively connected to each other, and the retaining portion **13**, composed of an aluminum alloy, and the lead portion **15** are conductively connected to each other, the ground potentials of the mating connector **100** and the communication connector **1** are kept at the same ground potential as the circuit board **200**.

[0039] The communication connector **1** is conductively connected to the enclosure **300** by pressing the ground piece **40**A against the enclosure **300**. Each of the two branches of the shielding portion **43**A of the ground piece **40**A and the enclosure **300** are conductively connected to each other, and thereby a plurality of grounded electric current paths can be formed. By forming a plurality of grounded electric current paths, the number of paths for releasing electric current that will cause noise to a portion where communication signals are not affected can be increased, and the occurrence of noise can be reduced.

[0040] When the communication connector **1**, having the lead portion **15** conductively connected to the through-hole portion of the circuit board **200**, is mated with the mating connector **100**, the shield member **110** of the mating connector **100** and the tubular member **30** are conductively connected to each other. In addition, the tubular member **30** and the retaining portion **13** retaining the tubular member **30** are conductively connected to each other, and the retaining portion **13**, composed of an aluminum alloy, and the attachment portion **20**A are conductively connected to each other. Since the attachment portion **20**A and the ground piece **40**A are conductively connected to each other, the ground potentials of the mating connector **100** and the communication connector **1** are kept at the same ground potential as the enclosure **300**.

[0041] By arranging the ground piece **40**A between the housing **10** and the enclosure **300**, in a projected part of the ground piece **40**A when the housing **10** is viewed from the front, the advance of radiation noise entering through the opening portion **301** between the housing **10** and the enclosure **300** can be shielded. This can reduce the effect of the radiation noise on the communication connector **1** and the circuit board **200**. It should be noted that "when the housing **10** is viewed from the front" refers to a view in the first direction X where the housing **10** is viewed from the front from the rear in the first direction X.
[0042] In the present embodiment, the lead portion **15** serves as a mount portion **15**A through-hole mounted on the circuit board **200**, but the lead portion **15** is not limited to this. For example, like the third contact section **51**_c_, the lead portion **15** may be a mount portion surface-mounted on a contact portion of the circuit board **200**.
[0043] Advantageous effects of the communication connector **1** according to FIGS. **1-5** will now be described.
[0044] The communication connector **1** of the present embodiment includes the housing **10** composed of an electrically conductive material, the tubular member **30** as the first conducting portion that is retained while being conductively connected to the housing **10**, and that is conductively connected to the shield member **110** of the mating connector **100**, the ground piece **40**A as the second conducting portion that is provided on the upper face **10**U as one face side of the housing **10**, and that is conductively connected to the housing **10**, and the lead portion **15** as the third conducting portion that is provided on the lower face **10**D as an other face side of the housing **10**, and that is conductively connected to the housing **10**.
[0045] The communication connector **1** comprises the housing **10** composed of an electrically conductive material. Therefore, for example, if the ground piece **40**A, as the second conducting portion, is brought into electrical contact with the enclosure **300**, and the lead portion **15**, as the third conducting portion, is electrically connected to the circuit board **200**, the enclosure **300** and the circuit board **200** are conductively connected to each other; thus, the advantageous effect of enabling the ground potentials of the mating connector **100** and the communication connector **1** to be kept at the same ground potential as the enclosure **300** can be achieved. In addition, by forming the plurality of grounded electric current paths, the number of paths for releasing electric current that will cause noise to a portion where communication signals are not affected can be increased, and the occurrence of noise can be reduced. Therefore, the communication connector **1** has a simple structure and yet can reduce the effect of electromagnetic noise.
[0046] An exemplary embodiment of a communication connector **1** according to FIGS. **6-7** will now be described.
[0047] In the communication connector **1** according to FIGS. **1-5**, the attachment portion **20**A is formed at the rear of the housing **10**, and the shielding portion **43**A of the ground piece **40**A is arranged in a position far from the opening portion **301**, whereas the communication connector **1** according to FIGS. **6-7** is different in that an attachment portion **20**B is formed at the front face **10**F of the housing **10**, and a shielding portion of a ground piece **40**B is arranged in a position close to the opening portion **301**. It should be noted that the same elements as those of the embodiment of the communication connector **1** according to FIGS. **1-5** are denoted by the same reference signs as in FIGS. **1-5** for the communication connector **1** according to FIGS. **6-7** and the descriptions thereof are omitted.
[0048] As shown in FIGS. **6-7**, the attachment portion **20**B is formed with triangular ribs **22**B protruding in the upward direction U from the upper face **10**U of the housing **10**, and connecting from side faces of the housing **10** on opposite sides in the second direction Y. As shown in FIG. **6**, at a middle section in the second direction Y of the attachment portion **20**B, a latch portion **21**B protruding from a face on the rear side is formed.
[0049] As shown in FIGS. **6-7**, the attachment portion **20**B is formed closer to the front face **10**F than the midpoint of a length of the housing **10** in the first direction X. That is, the attachment

portion **20**B is formed in the housing **10** at the mating opening **11** side where the mating connector **100** is mated.
[0050] As shown in FIG. **7**, the ground piece **40**B, in a side view, is formed in an inverted U-shape **41**B, a curved portion **48**B is formed in a front section **42**B of the U-shape **41**B, a shielding portion **43**B extending from the curved portion **48**B is formed, and a curved portion **44**B is formed on a distal end side of the shielding portion **43**B. A rear section **47**B of the U-shape **41**B, as shown in FIG. **7**, extends in the downward direction D, and extends obliquely downward in the rearward direction B on a distal end side.
[0051] As shown in FIG. **6**, the ground piece **40**B, in a top view, is formed with a notched portion **45**B in the shielding portion **43**B, and is divided into two branches. In addition, as shown in FIGS. **6-7**, an opening **46**B is formed in a middle section of the U-shape **41**B, and when the ground piece **40**B is attached to the attachment portion **20**B, the latch portion **21**B of the attachment portion **20**B enters the opening **46**B to prevent unexpected extraction of the ground piece **40**B.
[0052] As shown in FIG. **7**, the ground piece **40**B is arranged at the front face **10**F of the housing **10** between the housing **10** and the enclosure **300**, and a portion connecting from the curved portion **48**B to the shielding portion **43**B and the shielding portion **43**B elastically deform, thereby causing the curved portion **44**B to press a wall **310**F on the front side of the enclosure **300** in the first direction X. In a projected part of the ground piece **40**B, when the housing **10** is viewed from the front, radiation noise entering through the opening portion **301** between the housing **10** and the enclosure **300** can be blocked in a position at the front face **10**F of the housing **10**. This prevents the radiation noise from entering between the communication connector **1** and the enclosure **300**, thereby enabling a further reduction in the effect on the circuit board **200**.
[0053] An exemplary embodiment of a communication connector **1** according to FIGS. **8-9** will now be described.
[0054] In the communication connector **1** according to FIGS. **1-5** and the communication connector **1** according to FIGS. **6-7**, the shielding portion **43**A of the ground piece **40**A and the shielding portion **43**B of the ground piece **40**B are divided into two branches, whereas the communication connector **1** according to FIGS. **8-9** is different in that a shielding portion **43**C of a ground piece **40**C is continuous in the second direction Y. It should be noted that the same elements as those of the embodiment of the communication connector **1** according to FIGS. **1-5** are denoted by the same reference signs as in FIGS. **1-5** for the communication connector **1** according to FIGS. **8-9** and the descriptions thereof are omitted.
[0055] As shown in FIGS. **8-9**, the ground piece **40**C, in a side view, is formed in a U-shape **41**C lying on its side at the rear, an upper section **42**C of the U-shape **41**C is formed, a curved portion **48**C extending from the upper section **42**C and folded back is formed, a shielding portion **43**C extending from the curved portion **48**C is formed, and a curved portion **44**C is formed on a distal end side of the shielding portion **43**C. A lower section **47**C of the U-shape **41**C, as shown in FIG. **8**, extends in the frontward direction F, and extends obliquely downward in the frontward direction F on a distal end side.
[0056] As shown in FIG. **8**, in the present embodiment, when the housing **10** is viewed from the front face **10**F, that is, when the housing **10** is viewed in the first direction X, the projected shape of the ground piece **40**C is continuous in the second direction Y. A large shielding face for shielding radiation noise entering through the opening portion **301** between the housing **10** and the enclosure **300** can be formed. In the present embodiment, the dimension in the second direction Y of the ground piece **40**C is greater than the dimension in the second direction Y of the shield member **110** of the mating connector **100**. By making the dimension in the second direction Y of the ground piece **40**C greater than the dimension in the second direction Y of the shield member **110** of the mating connector **100**, the effect of the radiation noise on the communication connector **1** and the circuit board **200** can be reduced.
[0057] As shown in FIGS. **8-9**, since the shielding portion **43**C of the ground piece **40**C is

continuous in the second direction Y, the section modulus in the thickness direction of the shielding portion **43**C is greater than those of the shielding portion **43**A of the ground piece **40**A and the shielding portion **43**B of the ground piece **40**B that are divided into two branches. A pressing force onto the enclosure **300** can be secured by the shielding portion **43**C of the ground piece **40**C, even with a small amount of displacement of elastic deformation of the shielding portion **43**C as a flat spring, and the mechanical stability of the conducting portion can be improved.

[0058] An exemplary embodiment of a communication connector **1** according to FIGS. **10**A-**11** will now be described.

[0059] In the communication connector **1** according to FIGS. **6-7**, the attachment portion **20**B is formed closer to the front face **10**F than the midpoint of a length of the housing **10** in the first direction X, and the wall **310**F at the front side of the enclosure **300** in the first direction X is pressed by the curved portion **44**B of the ground piece **40**B. In contrast, as shown in FIG. **11**, the communication connector **1** according to FIGS. **10**A-**11** is different in that a ceiling face **320**U on the upper side of the enclosure **300** in the third direction Z is pressed by a curved portion **44**D of a ground piece **40**D. It should be noted that the same elements as those of the embodiment of the communication connector **1** according to FIGS. **6-7** are denoted by the same reference signs as in FIGS. **1-7** for the communication connector **1** according to FIGS. **10**A-**11** and the descriptions thereof are omitted.

[0060] As shown in FIGS. **10**A-**11**, the ground piece **40**D, in a side view, is formed in an inverted U-shape **41**D, and an upstream portion **42**D and a downstream portion **47**D that connect from the U-shape **41**D are formed. In a front view, an extended portion **43**D extending from the right toward the left and extending obliquely upward is formed, and a curved portion **44**D is formed on a distal end side of the extended portion **43**D.

[0061] As shown in FIG. **11**, in the present embodiment, the ground piece **40**D is attached to the attachment portion **20**B formed at the mating opening **11** side where the mating connector **100** is mated, which is the front face **10**F of the housing **10**. In addition, as shown in FIG. **12**, the ceiling face **320**U on the upper side of the enclosure **300** in the third direction Z is pressed by the curved portion **44**D of the ground piece **40**D. By forming a grounded electric current path at the front face **10**F of the housing **10**, and between the housing **10** and the ceiling face **320**U, the electric current conduction distance is shortened, so that the occurrence of electromagnetic noise can be reduced.

[0062] An exemplary embodiment of a communication connector **1** according to FIG. **12** will now be described.

[0063] In all of the embodiments of the aforementioned communication connectors **1** according to FIGS. **1-11**, the ground pieces **40**A, **40**B, **40**C, **40**D are attached to the attachment portions **20**A, **20**B, formed in the specific shapes, whereas, in the communication connector **1** according to FIG. **12** is different in that the shape for attachment of a ground piece **40**E is not limited to a specific shape, but the ground piece **40**E is installed on the upper face **10**U of the housing **10**.

[0064] As shown in FIG. **12**, the ground piece **40**E is installed on the upper face **10**U of the housing **10**. The housing **10** and the ground piece **40**E are bonded and secured to each other, for example, with an electrically conductive double-sided tape or the like. In an alternative embodiment, for example, the housing **10** and the ground piece **40**E are soldered to each other.

[0065] In the present embodiment, the enclosure **300** and the housing **10** can be electrically connected to each other with the ground piece **40**E attached to the upper face **10**U of the housing **10**. Since the housing **10** is electrically conductive, the ground piece **40**E can be attached in a desired position on the upper face **10**U of the housing **10**. By attaching the ground piece **40**E in a desired position on the upper face **10**U of the housing **10**, an electric current path can be formed in a desired position, the degree of freedom of countermeasures against the electromagnetic noise increases, and it becomes easier to take countermeasures against the electromagnetic noise.

[0066] Without departing from the spirit of the present invention, the configurations shown and described in the various exemplary embodiments described above may be selectively adopted or eliminated, or may be appropriately modified to other configurations.

Claims

1. A communication connector, comprising: a housing composed of an electrically conductive material and receiving a mating connector; a first conducting portion retained in the housing and conductively connected to the housing, the first conducting portion is conductively connected to a shield member of the mating connector; a second conducting portion provided on a side of the housing and conductively connected to the housing; and a third conducting portion provided on another side of the housing and conductively connected to the housing.
2. The communication connector of claim 1, wherein the first conducting portion has a tubular shape and is conductively connected to the shield member on an outer peripheral face of the tubular shape, and the second conducting portion elastically deforms when conductively connected to an enclosure.
3. The communication connector of claim 1, wherein the third conducting portion is a mount portion mounting the communication connector on a circuit board.
4. The communication connector of claim 2, wherein the second conducting portion presses a wall on a front face of the housing in a first direction of the enclosure and elastically deforms when conductively connected to the enclosure.
5. The communication connector of claim 1, wherein the second conducting portion is continuous in a second direction that intersects a first direction in which the mating connector is received in the housing, and a dimension of the second conducting portion in the second direction is greater than a dimension of the shield member of the mating connector in the second direction.
6. The communication connector of claim 1, wherein the second conducting portion is formed in a U-shape.
7. The communication connector of claim 6, wherein the second conducting portion has a shielding portion extending from an upper section of the U-shape.
8. The communication connector of claim 6, wherein the U-shape has a lower section extending in a frontward direction.
9. The communication connector of claim 6, wherein the U-shape has an opening in a middle section of the U-shape.
10. The communication connector of claim 7, wherein the shielding portion has a notched portion separating the shielding portion into two sections.
11. The communication connector of claim 1, further comprising a contact molded body disposed in the first conducting portion.
12. The communication connector of claim 11, wherein the contact molded body has a contact portion that contacts a mating contact portion of the mating connector.
13. The communication connector of claim 12, wherein the contact portion has a first contact section contacting the mating contact portion, a second contact section extending perpendicularly from the first contact section, and a third contact section extending from the second contact section.
14. The communication connector of claim 13, wherein the third contact section extends out from the housing and is surface-mounted on a circuit board.
15. The communication connector of claim 1, wherein the second conducting portion is attached to an attachment portion of the housing.
16. The communication connector of claim 15, wherein a latch portion of the attachment portion enters an opening of the second conducting portion.
17. The communication connector of claim 15, wherein the attachment portion has triangular ribs protruding in an upward direction away from an upper face of the housing.

18. The communication connector of claim 1, further comprising a plurality of third conducting portions extending from a lower face of the housing.